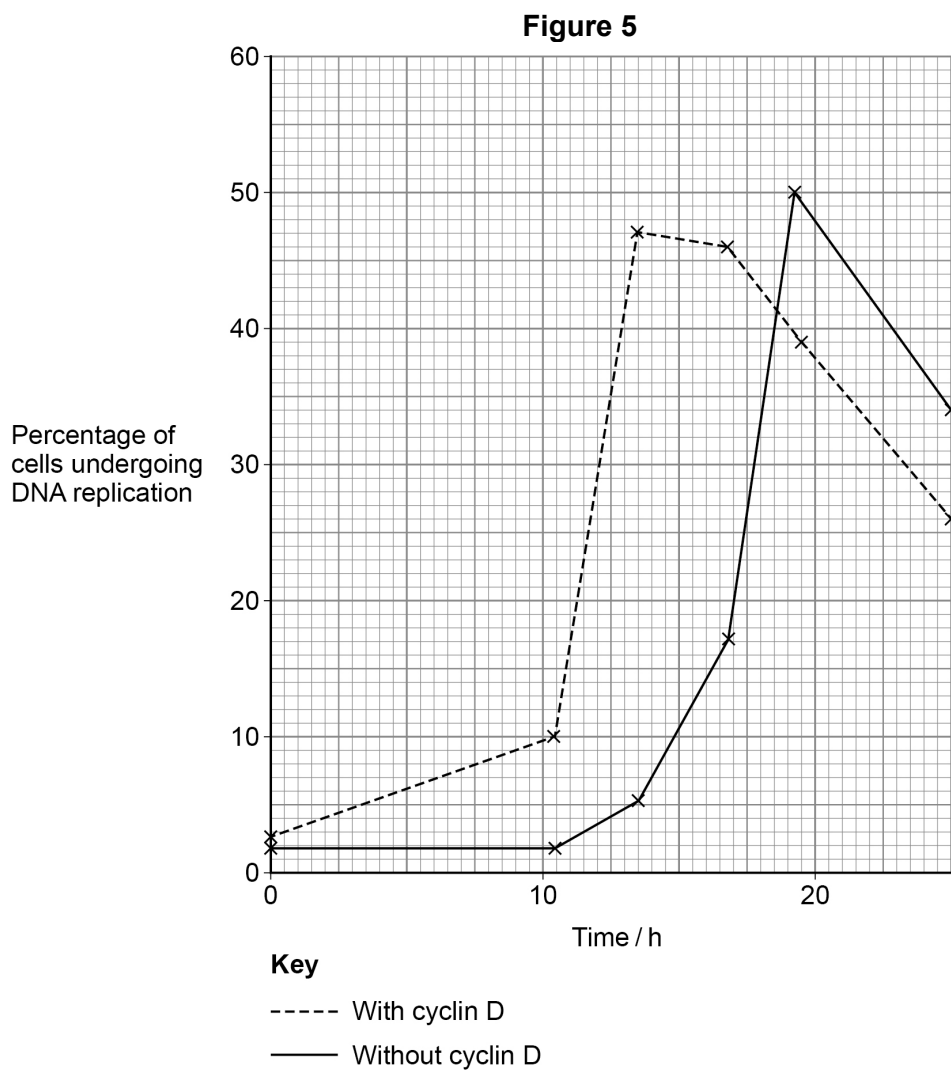


0 5 . 1

Describe the role of DNA polymerase in the semi-conservative replication of DNA.

[2 marks]

Figure 5 shows the percentage of rat cells undergoing DNA replication. Some cells contained a protein called cyclin D and some cells did not contain cyclin D. All cells were in early interphase at time 0



0 5 . 2

It took less time for 25% of cells with cyclin D to be undergoing DNA replication than for 25% of cells without cyclin D.

Use **Figure 5** to calculate this time difference as a percentage decrease.

Show your working.

[2 marks]

Answer _____ %

0 5 . 3

Cyclin D stimulates the phosphorylation of DNA polymerase, which activates the DNA polymerase.

Describe how an enzyme can be phosphorylated.

[2 marks]

0 5 . 4

Some tumour cells contain higher than normal concentrations of cyclin D.

Use **Figure 5** to suggest why higher than normal concentrations of cyclin D could result in a tumour.

[2 marks]
